**Hematology Consultation**

**CONSULTING PHYSICIAN:** Dr. Eleanor Vance, MD

**HISTORY:**
I was asked to see Mr. Rogers, a 66-year-old male with a history of relapsed AML, to provide recommendations for salvage therapy. I have followed Mr. Rogers since his initial diagnosis.

To summarize his course:
He was diagnosed with AML on December 7, 2021. His diagnostic marrow showed **trisomy 8 (+8)** as the sole cytogenetic abnormality. This is not classified as favorable or adverse, thus placing him in the **ELN 2022 Intermediate-risk** category. He was treated with standard "7+3" induction chemotherapy since mid December, achieved a complete remission, and subsequently completed three cycles of HiDAC consolidation, finishing in July 2022.

He enjoyed an excellent remission. Unfortunately, his routine surveillance labs in September 2023 showed declining counts. A bone marrow biopsy on September 12, 2023, confirmed a frank relapse, with 60% blasts that are morphologically and immunophenotypically identical to his original leukemia.

**CURRENT STATUS:**
Mr. Rogers is currently an inpatient for management of symptomatic pancytopenia (Hgb 7.5, Plt 22). His performance status is good (ECOG 1). He is physically active and motivated for further therapy with curative intent. His comorbidities include well-controlled hypertension and a remote history of a pulmonary embolism in 2010 (no longer on anticoagulation). His renal and hepatic function are normal. Cardiac evaluation shows a preserved LVEF of 55%.

**ASSESSMENT & OPTIONS:**
This is a first relapse of intermediate-risk AML with a relatively long duration of first remission (CR1 > 12 months). This is a favorable prognostic factor for response to salvage chemotherapy. The primary goal of salvage therapy would be to induce a second complete remission (CR2) to serve as a bridge to a potentially curative allogeneic stem cell transplant.

I have discussed the following options in detail with Mr. Rogers and his wife:

**1. Intensive Salvage Chemotherapy:**

- **Regimen:** FLAG-Ida (Fludarabine, Cytarabine, Idarubicin, G-CSF).
- **Pros:** High rates of achieving CR2 in patients with a long CR1. It is a well-established standard of care.
- **Cons:** Highly myelosuppressive and toxic. Involves a 4-6 week hospitalization with a near-certainty of febrile neutropenia and other complications. There is a treatment-related mortality risk of 5-10%.

**2. Venetoclax-Based Salvage Therapy:**

- **Regimen:** Venetoclax in combination with a hypomethylating agent (e.g., Azacitidine) or low-dose Cytarabine.
- **Pros:** Generally better tolerated than intensive chemotherapy. Can often be administered primarily as an outpatient. Good efficacy data in the relapsed/refractory setting.

- **Cons:** Less likely to produce the deep, MRD-negative remission that is ideal prior to transplant. Response may be slower to emerge.

**3. Clinical Trial:**

- **Pros:** Offers access to novel agents and mechanisms of action (e.g., menin inhibitors, antibody-drug conjugates).
- **Cons:** We currently have an open trial for a CD123-targeted agent for which he would be eligible. However, efficacy is investigational, and toxicities may be unknown.

**RECOMMENDATION & PLAN:**
After a long discussion, Mr. Rogers continues to express a strong desire to pursue the most aggressive, curative-intent option. He understands the significant risks associated with intensive chemotherapy but feels that the potential benefit of bridging to a curative transplant is worth those risks. His good performance status and organ function make him a suitable candidate.

Therefore, my primary recommendation is to proceed with **intensive salvage chemotherapy with FLAG-Ida as a bridge to allogeneic stem cell transplantation from haploidentical related donor.**

**The plan is as follows:**

1. Complete allogeneic stem cell transplant workup immediately: schedule high-resolution HLA typing for him and his two siblings. Consult the BMT service concurrently.
2. Once the infectious disease workup is clear, place a new double-lumen central venous catheter.
3. Initiate the FLAG-Ida regimen, targeting the start of next week.
4. Aggressive supportive care with prophylactic antimicrobials and transfusion support as needed.
5. Perform a bone marrow biopsy around Day +28 to assess for CR2. If remission is achieved, we will proceed directly to transplant conditioning.

Mr. Rogers and his wife voiced understanding and agreement with this plan. They are aware of the arduous path ahead but remain hopeful.

**DATE:** 18-Sep-2023
**PATIENT:** ROGERS, Walter (DOB 24-Apr-1957)
**MRN:** AML057
**REASON FOR CONSULT:** Relapsed Acute Myeloid Leukemia; Discussion of Second-Line Therapy Options.

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**Dr. Eleanor Vance, MD**
Hematology-Oncology Service